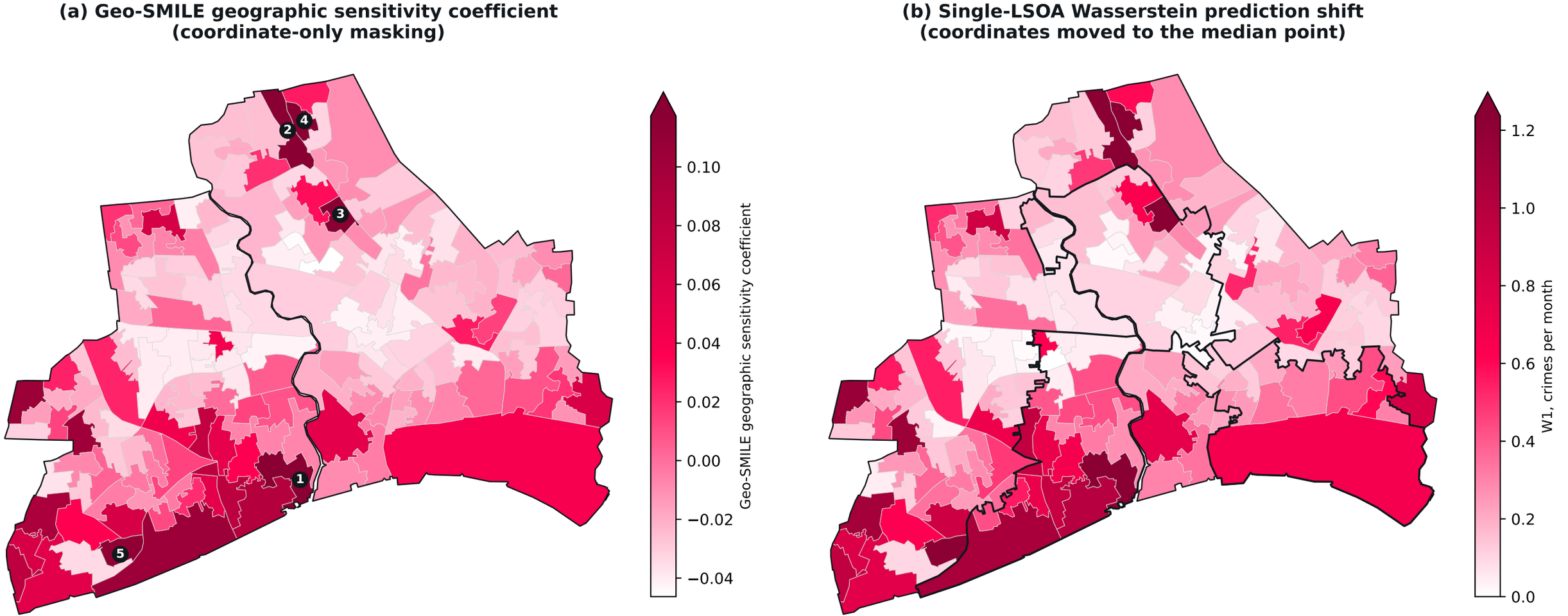


Coordinate-only design (as Seattle/Chicago): the XGBoost model uses the 30 predictors plus centroid longitude/latitude; masking an LSOA moves only its coordinates to the training-median point, all other inputs kept. N=4,000, seed 42. Numbers mark the top 5 by signed coefficient. Panel (b): black outlines = expert-highlighted regions (58 LSOAs). One sequential scale per panel (darker = more sensitive), capped at the 98th percentile; the sign is not a direction of prediction change. Not causal effects.